Lokeswarreddy Basireddy

Computer Science Undergraduate | Pre-final year
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Professional Summary

Computer Science Undergraduate specializing in Cybersecurity and Artificial Intelligence, with practical experience in Secure Application Development and AI-Powered Security Systems. Experienced in building Real-Time Proctoring Solutions and Encryption-Based Tools to enhance Digital Trust and System Integrity. Committed to developing Ethical, Scalable, and Secure Technology Solutions.

Experience

AI Automation Intern — Tech Mahindra
Project Name : Neo Hire (CIO Automation Initiative)

- Developed a real-time AI-based proctoring system using computer vision to detect suspicious activities during online interviews.
- Built and optimized live webcam processing pipelines (15–25 FPS) for low-latency behavior monitoring.
- Implemented automated event detection with timestamp logging and structured CSV report generation for audit and review.
- Integrated backend services using Flask (Python) to manage detection workflows and reporting.

Tech Stack: Python, OpenCV, Flask, Real-Time Video Processing, Automation

Projects

File Encryption & Decryption Tool

Tech Stack: Python, AES-256-GCM, Cryptography Library, Secure Key Management), Git

- Developed a Python-based file encryption and decryption tool using **AES-256-GCM**.
- Ensured **data confidentiality and integrity** through authenticated encryption.
- Implemented secure **password-based key management**.
- Resolved key-change issue by **re-wrapping the master key instead of re-encrypting files**, improving efficiency.

Smart Agriculture Web App

Tech Stack: React, Supabase (Edge Functions), Google API, Gemini API

- Developed a web platform with **role-based access control** for farmers and vendors, including secure **email authentication** via Supabase.
- Implemented **location-based features** to connect farmers with nearby vendors and enable real-time equipment booking.
- Integrated **real-time API calls** to fetch mandi prices from the **Government of India**, providing up-to-date market insights.

Core Skills

Languages: Python, C, HTML, CSS, JavaScript, Shell Script
Tools & Frameworks: Kali Linux, Wireshark, Nmap, NumPy, BigQuery ML
Concepts: Cybersecurity, DSA , Machine Learning, Cryptographic Algorithms
Platforms: Google Cloud, Kali Linux,

Certificates

- Security Operations Center in Practice - **IBM**
- Python for Data Science, AI & Development - **IBM**
- Develop GenAI Apps with Gemini and Streamlit - **Google**
- Inspect Rich Documents with Gemini Multimodality and Multimodal RAG - **Google**

Achievements

- Selected as a **Grand Finale Participant** at Google – Hack2Skill GenAI Hackathon for building scalable Generative AI solutions
- Led a team in the Smart Kissan AI Hackathon, designing AI-based agricultural decision-support solutions.
- Competed in CTF (Marvo), solving real-world cybersecurity challenges across web, crypto, and network domains.